Branch Coverage Report

Supply Chain Management Simulator

Metric	Result
Test cases collected	393
Tests passed	393
Tests failed	0
Total statements	922
Missed statements	18
Total branches	410
Partial branches	18
Overall coverage	97%
Coverage requirement	**80%**
Result	PASSED

1. Objective

This report evaluates the statement and branch coverage of the Supply Chain Management Simulator source code. Branch coverage helps identify decision paths that are not fully exercised by the automated test suite.

2. Test Environment and Execution

Platform: Windows (win32)
Python version: 3.12.7
Pytest version: 8.3.3
Coverage plugin: pytest-cov 7.1.0
Test framework: pytest
Total execution time: 3.92 seconds

Command executed:

```
python -m pytest --cov=src --cov-branch --cov-report=term-missing
--cov-report=html:docs/reports/coverage/branch
```

3. Module-wise Coverage

Module	Statements	Missed	Branches	Partial	Coverage
__init__.py	0	0	0	0	100%
clock.py	39	0	16	0	100%
config_loader.py	146	0	96	0	100%
costs.py	37	0	4	0	100%
demand.py	46	3	22	1	94%
forecasting.py	48	0	18	0	100%
inventory.py	77	6	36	6	89%
kpis.py	89	0	58	1	99%
main.py	0	0	0	0	100%
orders.py	72	0	26	0	100%

Module	Statements	Missed	Branches	Partial	Coverage
products.py	62	0	30	0	100%
purchase_orders.py	68	1	18	1	98%
reports.py	0	0	0	0	100%
shipments.py	0	0	0	0	100%
simulation.py	142	8	42	9	91%
suppliers.py	96	0	44	0	100%

4. Missing Coverage Details

The following executable lines were reported as missing by the coverage tool:

demand.py: 13-14, 62

inventory.py: 69, 72, 78, 88, 135, 140

kpis.py: 226 -> 223

purchase_orders.py: 74

simulation.py: 42, 63, 70, 73, 84, 102, 193 -> 196, 256, 298, 338 -> 343

The partial branch count indicates decision paths for which coverage.py observed only one outcome. These areas may benefit from additional tests if the corresponding paths are important to the expected application behaviour.

5. Analysis

The project achieved 97% overall coverage, exceeding the required 80% coverage threshold by 17 percentage points. All 393 collected tests passed successfully.

Several modules achieved 100% coverage. The lowest reported module coverage was inventory.py at 89%, followed by simulation.py at 91%. These modules contain the largest reported opportunities for additional coverage.

6. Conclusion

The automated test suite passed all 393 tests and achieved 97% overall coverage. The project therefore satisfies the specified 80% coverage requirement. The generated HTML report provides detailed, line-level coverage information for further inspection.

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